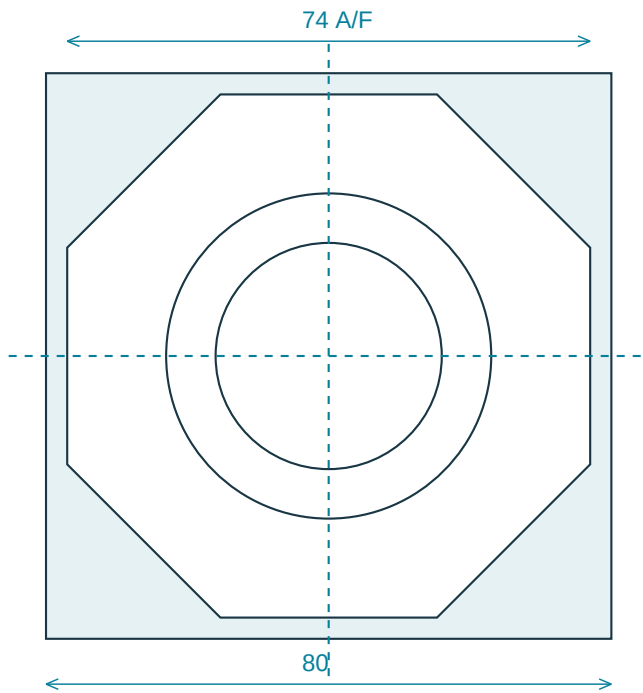


OCTAGONAL INSTRUMENT HOUSING

UMC-08 / REV A PROJECT-AUTHORED BENCHMARK - NOT A CUSTOMER DRAWING

01 TOP VIEW



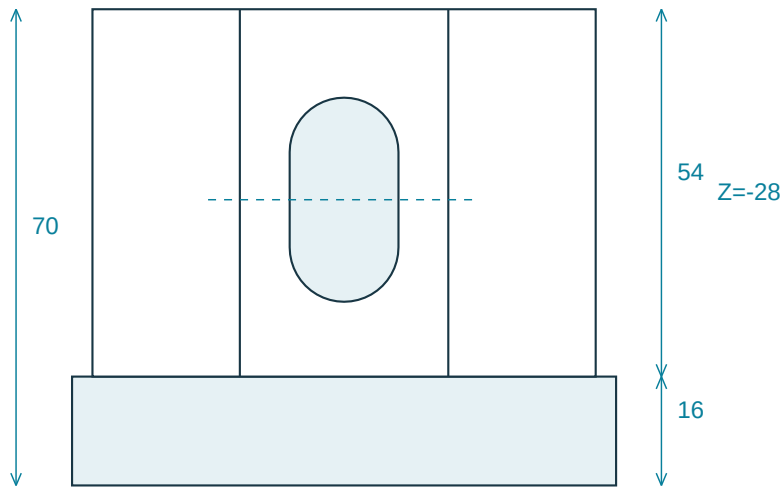
Cavity: DIA 46 x 5; DIA 32 x 18 TOTAL

Concentric at X0 Y0; depths from Z0.

8 equal facets; normals at 0,45,...315 deg.

Flange outer boundary remains 80 x 80.

02 TYPICAL FACET / ELEVATION



8X BLIND VERTICAL OBOUND

Width 16, height 30, end radius R8.

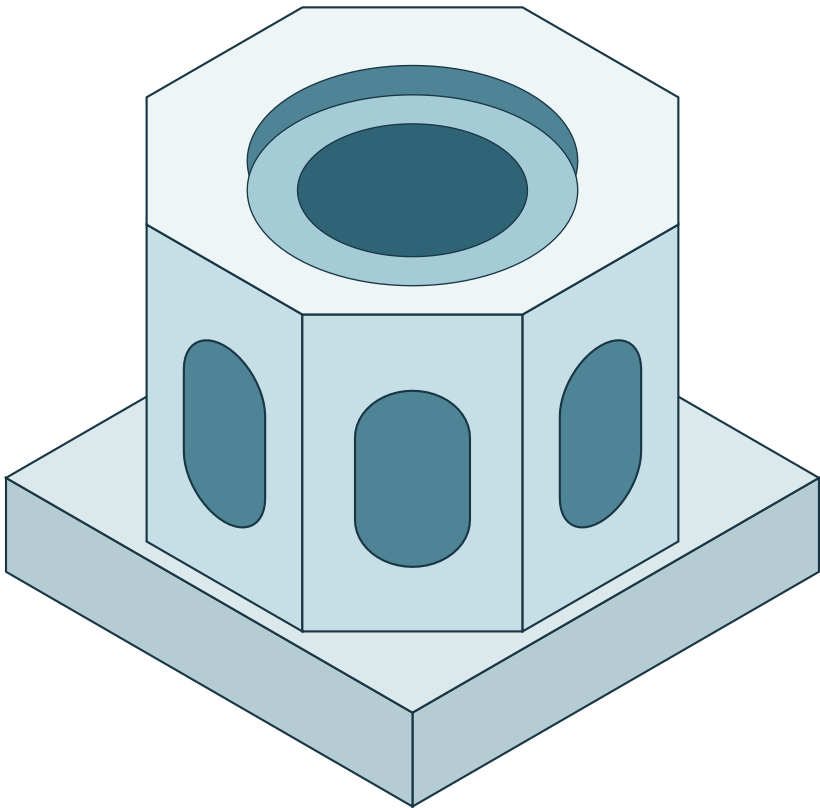
Centered on facet: U0, Z=-28.

Depth 5 normal inward from finished facet.

Slot spans Z=-43 to -13; floor is planar.

No slot intersects top cavity or flange.

03 ANALYTICAL ISOMETRIC



Illustrative projection of the specified geometry.

Not CAD output, a simulation, or a completed part.

COMPLETE GEOMETRY / DATUM CONTRACT

All dimensions mm. Origin X0 Y0 Z0 = original blank top center; +Z up.

Base: X and Y in [-40,+40], Z in [-70,-54]. Base stays entirely solid.

Upper: Z in [-54,0], regular octagon across flats 74, centered X0 Y0.

Exact facet planes: $X \cos(\theta) + Y \sin(\theta) = 37$;

$\theta = 0, 45, 90, 135, 180, 225, 270, 315$ deg. Solid lies ≤ 37 for all planes.

Facet length = $74 \tan(22.5 \text{ deg}) = 30.6518$ REF. No facet-edge chamfers.

For each slot: $U = (-\sin(\theta), \cos(\theta), 0)$, origin on face at Z=-28.

Subtract its 16 x 30 obround profile inward 5 along the face normal.

Top cavity: circular DIA46 to Z=-5; concentric DIA32 to Z=-18.

All pocket floors flat. Sharp theoretical exterior edges; no added fillets.

MATERIAL / MANUFACTURING BOUNDARY

Material: AL6061. Start: pre-sized 80 x 80 x 70 blank.

General linear tolerance +/-0.127; angles +/-0.2 deg.

Machine: Haas UMC-750; indexed 3+2 access to eight facets.

Only T1 DIA12.7 flat endmill; flute 25.4, stickout 33.02.

Upper exterior machining IS required. Retain square lower base.

No bottom machining, threads, through holes or extra features.

Blank is attached from below to the existing tall pedestal.

Rigid mounting assumed. Tool/holder/fixture clearance must pass

the actual simulator. Drawing is NOT a manufacturing approval.

Edges deburred manually after machining; no modeled edge breaks.